MIL-P-36907A
29 September 1976
SUPERSEDING
MIL-P-36907 (DSA-DM)
2 November 1972

MILITARY SPECIFICATION

PIPET-DILUENT, BLOOD, LABORATORY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of pipet with diluent for collecting and diluting specimens for hemoglobin determination.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.

Military

MIL-P-116	Preservation-packaging, Methods of.
MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6630

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

FOOD AND DRUG ADMINISTRATION

Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be sent to the Food and Drug Administration, Washington, DC 20204.)

3. REQUIREMENTS

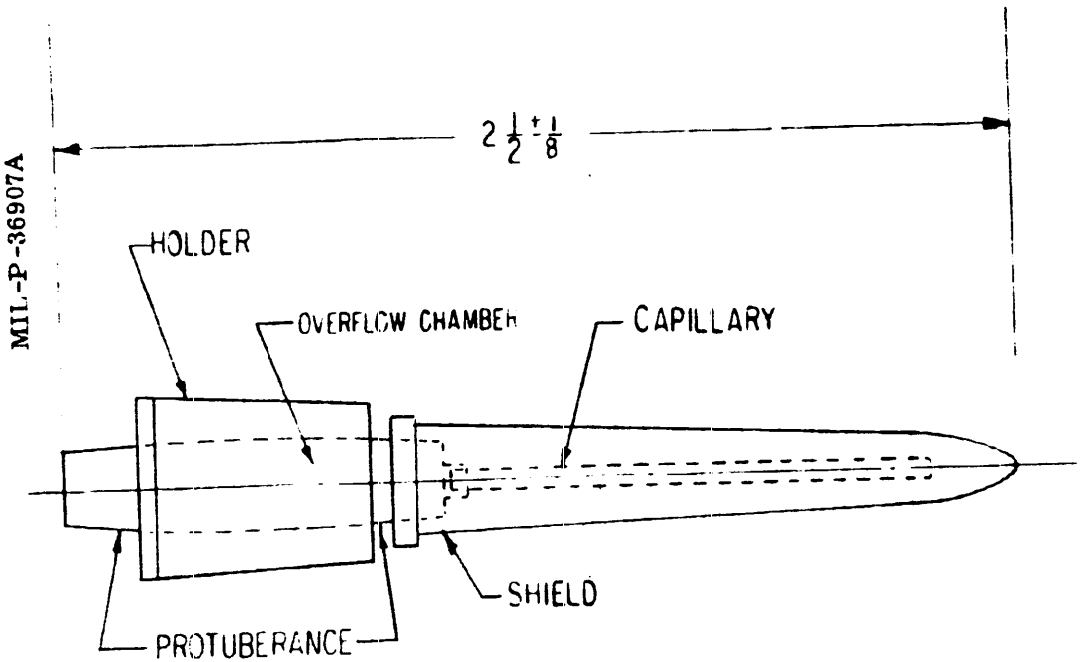
3.1 Material. Material shall be as specified herein.

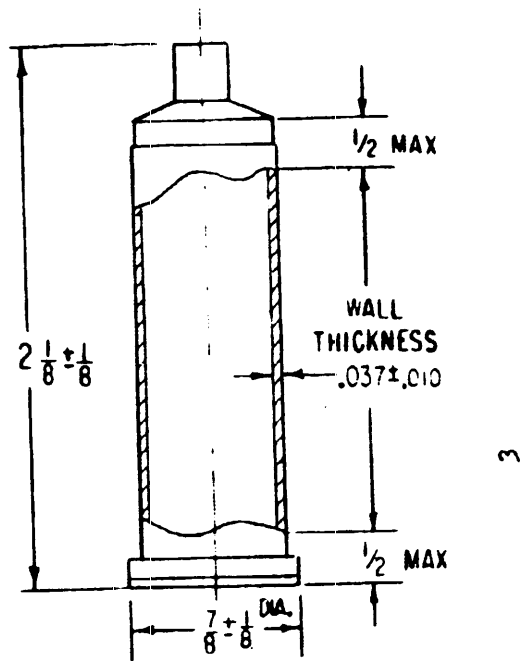
* 3.2 Construction. Pipet-diluent shall consist of a reservoir, holder, pipet (glass capillary tube) and a polypropylene capillary shield. Pipet-diluent shall be constructed in accordance with Figure 1. For this specification the glass capillary tube which functions as a pipet shall be designated a capillary tube.

3.2.1 Reservoir. Reservoir shall be fabricated of polyethylene. Top of reservoir shall have a diaphragm which can be pierced by the tip of the capillary shield. Reservoir shall contain an aqueous solution of stable cyanmethemoglobin reagent. Ethylene glycol shall be added to the solution to prevent freezing. Aqueous solution shall be free of particulation when examined macroscopically.

3.2.2 Holder. Holder shall be fabricated of polyethylene and shall secure the capillary tube. Shall have an overflow chamber at the end of the capillary

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RESERVOIR

FIGURE 1. Pipet-diluent, blood, laboratory

Dimensions are in inches.

METRIC CONVERSION TABLE

INCHES	MILLIMETERS (Approx)
.010	.3
.037	.9
1/8	3
1/2	13
7/8	22
2-1/8	54
2-1/2	63

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tube. Thickness of plastic shall be such that the holder is rigid. There shall be protuberances at each end. Both of the protuberances shall be tapered. The bottom protuberance shall make a snug fit with the neck of the reservoir and the top protuberance shall make a snug fit with the capillary shield.

* 3.3 Color coding. Holder shall be marked in yellow with "20 ul." (20 microliters).

3.4 Performance.

* 3.4.1 Volume. Reservoir shall contain 4.98 milliliters (ml.) ± 0.050 when tested as specified in 4.4.1.

3.4.2 Pipet (capillary tube) capacity. Capillary tube shall contain 20 ± 0.4 microliters (ul.) when tested as specified in 4.4.2.

3.4.3 Seal and capillary tube attachment. There shall be no evidence of leakage, and capillary tube shall not fall out when tested as specified in 4.4.3.

* 3.4.4 Hemoglobin analysis. The hemoglobin determination shall be identical to the standard ± 0.3 grams (g.) per 100 ml. when tested as specified in 4.4.4.

3.4.5 Hemolysis time. Solution shall be clear when tested as specified in 4.4.5.

3.5 Instructions. Each package shall contain at least one instruction sheet, which cites procedure for collecting and diluting blood. Marking on instruction sheet shall be legible and permanent.

* 3.6 Date of delivery. Date of delivery shall be not more than 60 days after date of manufacture.

* 3.7 Expiration date. The expiration date shall be not less than 18 months from the date of manufacture.

* 3.8 FDA requirements. Pipet-diluent shall be in accordance with Federal Food Drug and Cosmetic Act and CFR, Title 21, Part 328, In Vitro Products for Human Use.

3.9 Workmanship. Workmanship shall be first class throughout. Pipet-diluent shall be free of defects listed in Table III and other defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility ior inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Table I and MIL-STD-105. Unit of product shall be one collection and dilution unit.

Table I For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be in accordance with Table II and MIL-STD-105. Acceptance number shall be zero, and inspection level shall be S-1.

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Table II For tests

Unit of product	Characteristic	Requirement	Test procedure
Reservoir	Diluent volume	3.4.1	4.4.1
Holder	Pipet capacity	3.4.2	4.4.2
One collection and dilution unit	Seal	3.4.3	4.4.3
One collection and dilution unit	Hemoglobin analysis	3.4.4.	4.4.4
One collection and dilution unit	Hemolysis time	3.4.5	4.4.5

4.3 Examination. The pipet-diluent shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination of pipet-diluents shall be in accordance with Table III but shall not be limited to the defects listed therein.

Table III Classification of defects

Categories	Defects
Major	
101	Collection and dilution unit not free of dirt, grease or other contamination.
102	Collection and dilution unit not free of cracks and breaks.
103	Component missing.
104	Marking on holder not yellow.
105	Holder does not fit in reservoir.
106	Capillary tube broken.
107	Capillary shield not secured to holder.
108	Reservoir does not contain a liquid.
109	Liquid in reservoir not free of particles.

4.3.2 Dimensional examination. Pipet-diluent shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of the preservation-packaging and interior package markings shall be in accordance with the group A and B quality

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conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Volume of diluent. Reservoir with diluent shall be weighed. Reservoir shall be opened, emptied, cleaned and dried. Dry reservoir shall be weighed. Care shall be taken not to lose any plastic during opening, cleaning and drying procedures. Density of diluent shall be determined. Volume of diluent shall be calculated from the weight as corrected by the density.

4.4.2 Capillary tube capacity. Capacity of capillary tube shall be determined volumetrically or gravimetrically. If determined gravimetrically, capillary tube shall be removed from the holder. Capillary tube shall be weighed, filled with distilled water and reweighed. Weight of water shall be determined by subtracting first weighing from second weighing. Before the second weighing is made, tube should be observed to determine that there is no water on the outside. Volume in microliters shall be calculated by applying proper temperature corrections.

4.4.3 Seal and capillary tube attachment. Diaphragm at the top of the reservoir shall be pierced by tip of capillary shield. Capillary tube shall be inserted into reservoir and applicable protuberance shall be secured in the neck of the reservoir. Finger shall be placed over overflow opening. Unit shall be inverted and shaken by hand for one minute. Clearing of junction between protuberance and neck of reservoir or drops of liquid on the exterior of reservoir or holder shall be considered evidence of leakage.

4.4.4 Hemoglobin analysis. A curve which indicates hemoglobin content in grams percent (g. per 100 ml.) shall be prepared from photometric readings of standard hemoglobin/cyanmethemoglobin solutions at 540 nanometers. Curve shall have five points which indicate 24, 18, 12, 6 and 0 (blank) or 20, 15, 10, 5 and 0 (blank) g. per 100 ml. Either transmittance or absorbance readings may be used to describe curve.

Blood shall be collected and diluted in accordance with contractor's instructions, and dilution shall be viewed in a photometer at the wavelength of 540 nanometers. Hemoglobin content (g. per 100 ml.) shall be determined by marking the photometric reading on the previously mentioned curve.

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The preceding procedure shall be repeated using a sample of the same blood and a different pipet-diluent unit. There shall be proof that this pipet-diluent unit will produce a correct dilution (standard).

4.4.5 Hemolysis time. Blood shall be diluted in accordance with contractor's instructions and allowed to stand 10 minutes after being mixed.

5. PACKAGING.

* 5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package, containing 50 pipets with holders and 50 reservoirs as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Immediate containers. Fifty pipets with holders shall be packaged in a plastic bag and the bag shall be secured to prevent spilling of contents under normal handling. Fifty reservoirs shall be packaged in a glass jar containing sufficient moisture to maintain volume stability of the diluent. Closure of the jar shall be adequate to prevent spilling of the contents or loss of liquid.

* 5.1.1.1.2 Unit packaging. The bag of pipets and the jar of reservoirs shall be packaged with an instruction sheet in a corrugated box of appropriate size and design. The box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure shall be adequate to prevent accidental opening. Packaging shall conform to MIL-P-116, method III.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

* 5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

* 5.2.1 Level A. Ten units of issue shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. Ten units of issue shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Immediate containers. Each plastic bag containing pipets and holders need not be marked. Each jar containing the reservoirs shall bear the following information (not required to appear in the sequence indicated):

a. Statement that item is for hemoglobin determination.
b. Volume of cyanmethemoglobin diluents per reservoir, ingredients of cyanmethemoglobin diluent, statement that a 20 ul. pipet is used and that dilution ratio between volume of sample and volume of diluent is 1:250.
c. Statement that jar contains sufficient moisture to maintain volume stability of the diluent.
d. Statement that jar contains 50 reservoirs.
e. Lot and/or control numbers and expiration date.
f. Name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear. When both names are placed on the label, the following designations shall precede the names:

"MFR" for manufacturer and
"CONTR" for contractor.

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* 5.3.2 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. In addition, the markings as specified for the immediate containers (jars) shall also be shown. Marking shall also include statement that each package contains 50 pipets. In addition, Type I shelf life marking as specified in MIL-STD-129 shall be shown.

* 5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number shall be shown. In addition, Type I shelf life marking as specified in MIL-STD-129 shall be shown.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type normally procured to meet Military requirements.

6.3 This specification is applicable to the following item listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6630-00-004-1473	PIPET-DILUENT, BLOOD, LABORATORY, Plastic, Disposable, 20 Microliter Capacity, Hemoglobin Determination, 50s

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6630-0269

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DOCUMENT IDENTIFIER AND TITLE

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NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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